

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE: CSA1102

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture.
(BL3)

Assessment Tool Weightage: 60%

QUESTIONS: 6

Total Marks:60

Annexure A

Application Based Questions

Code of Conduct:

*I N. Hima Harshitha (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

N. Hima Harshitha

Annexure A

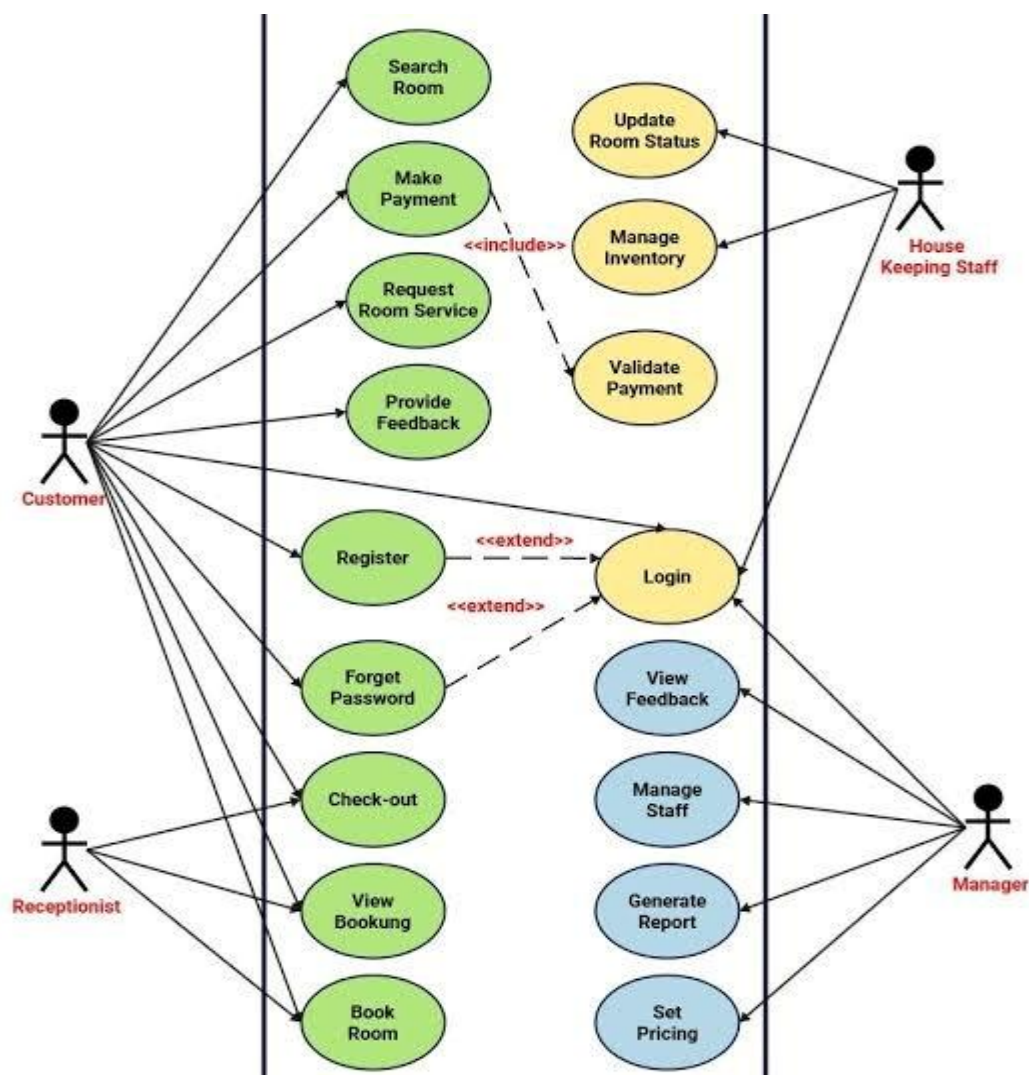
Application Based Questions

1. Use Case Diagram Analysis

A Smart Hotel Reservation System allows guests to search rooms, make reservations, cancel bookings, and process payments. Managers handle room allocation and reports, while payment services process transactions. The current design does not clearly distinguish actor responsibilities.

Analyse the system requirements, identify missing actors and use cases, and design an optimized Use Case Diagram showing proper relationships (include, extend, generalization) for better system clarity.

Smart Hotel Reservation System

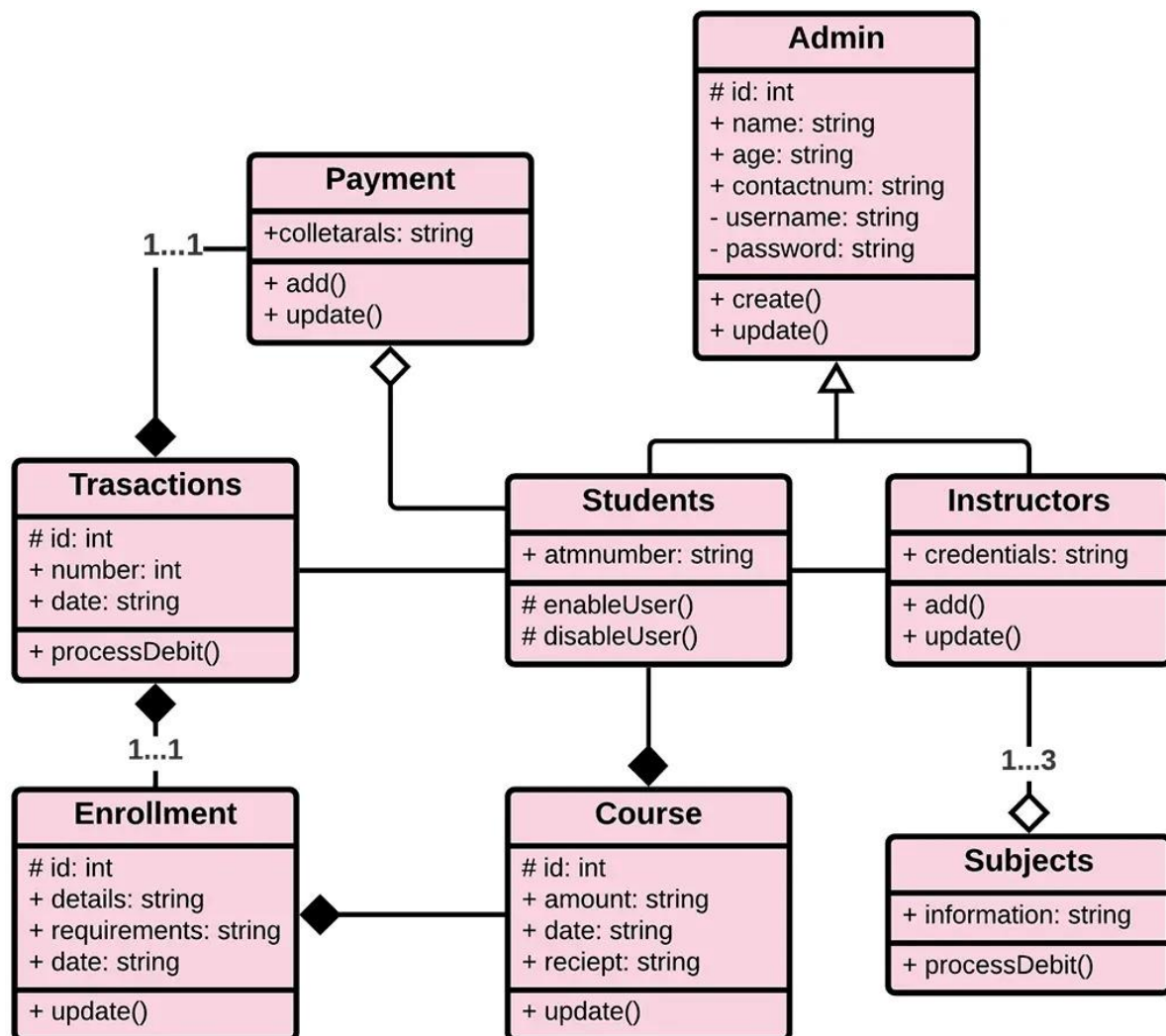


2. Class Diagram Design

A University Management System includes entities such as Student, Faculty, Course, Department, and Examination. The existing design lacks proper attributes, operations, and relationships among classes.

Analyse the system, identify relevant classes with attributes and methods, define relationships (association, aggregation, inheritance), and construct a well-structured Class Diagram.

University Management System

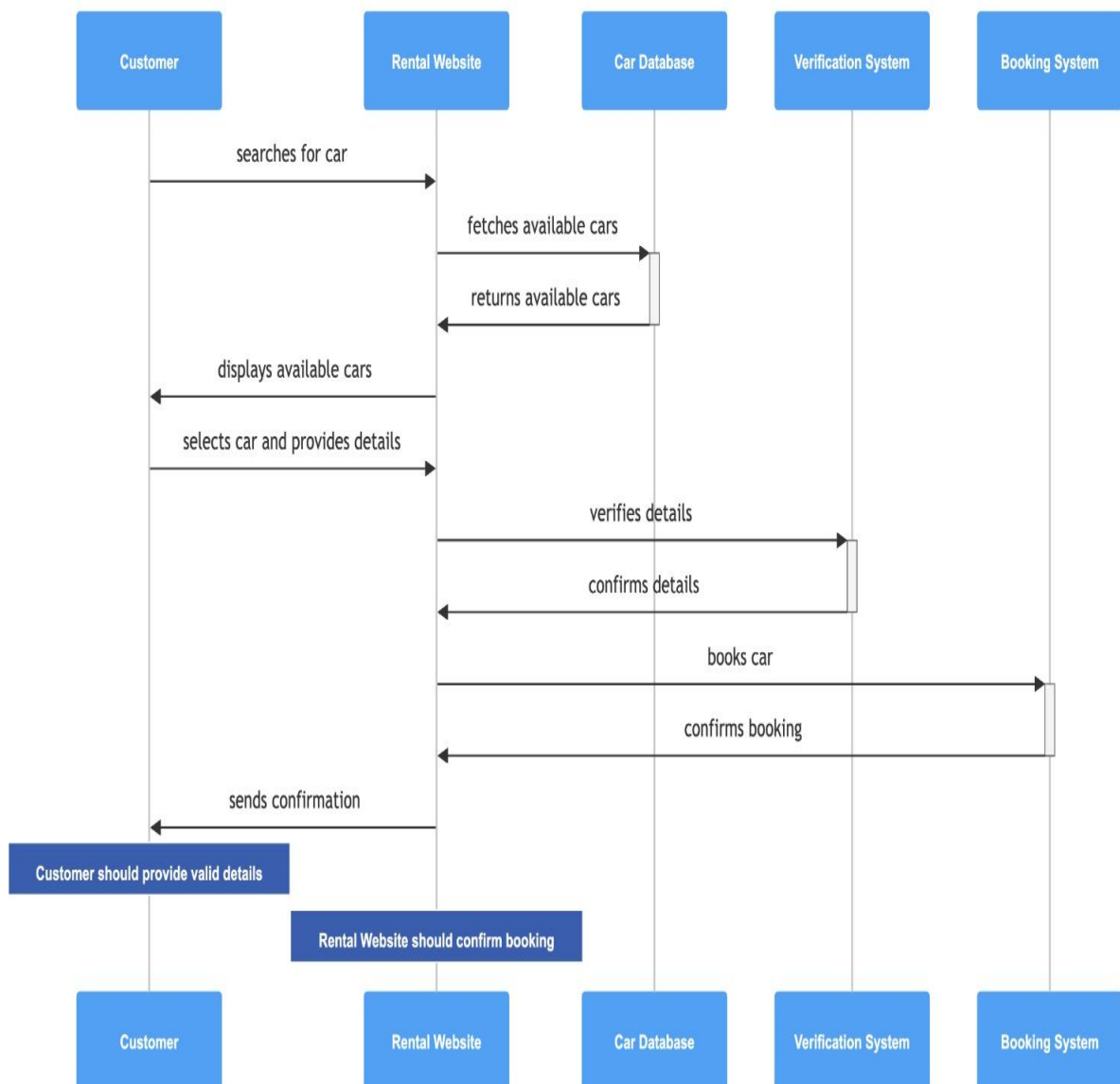


3. Sequence Diagram Evaluation

In a Vehicle Rental System, a customer searches for a vehicle, books it, makes payment, and receives a rental confirmation. The interaction among system components is incomplete and poorly documented.

Analyse the process flow, identify missing objects and message sequences, and design a detailed Sequence Diagram representing proper interaction among system components.

Vehicle Rental System

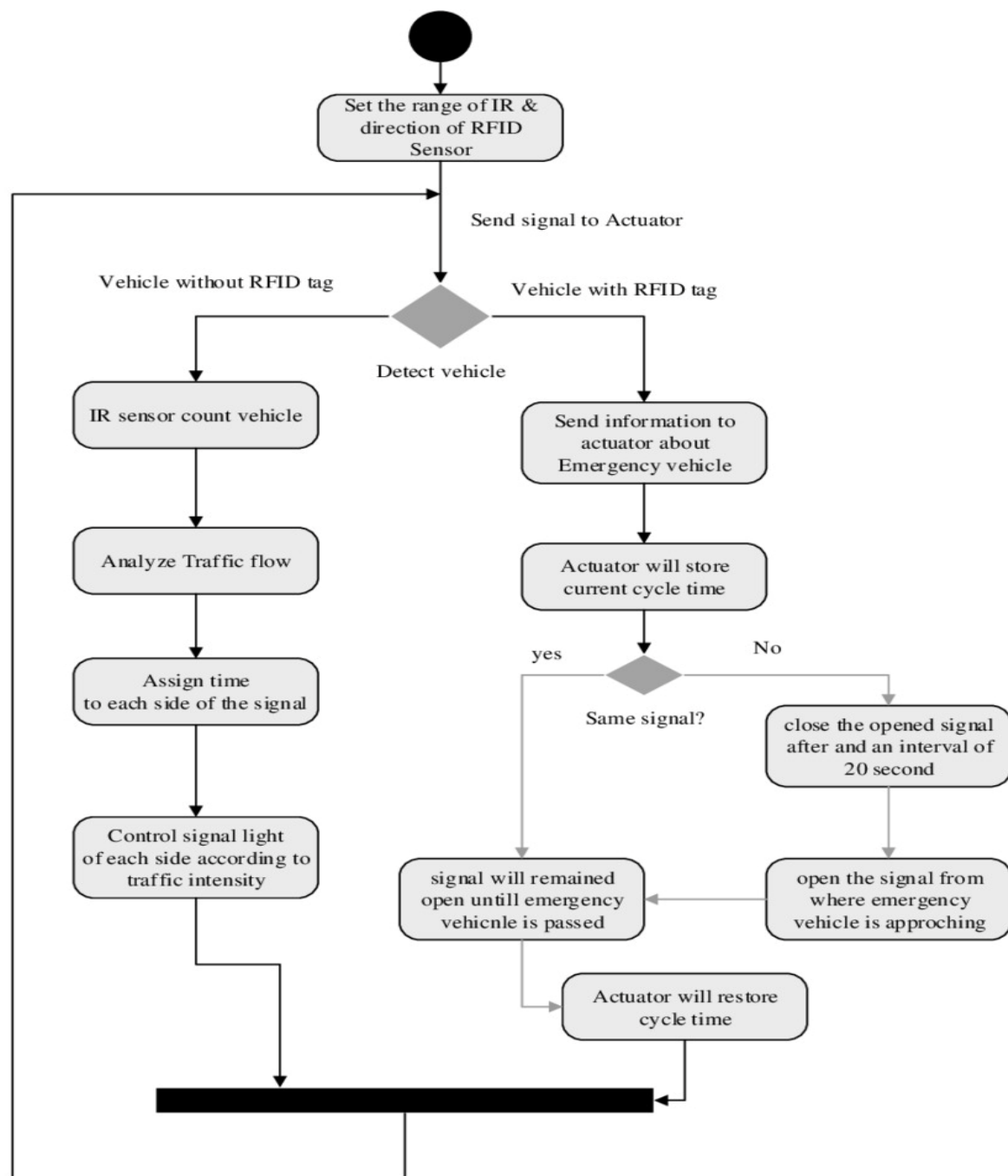


4. State Diagram Construction

A Traffic Signal Control System operates through states such as Red Signal, Green Signal, Yellow Signal, Emergency Mode, and Maintenance Mode. The transitions between states are not clearly defined.

Analyse the system behaviour, identify valid states and transitions, and construct a State Transition Diagram ensuring all events and conditions are properly represented.

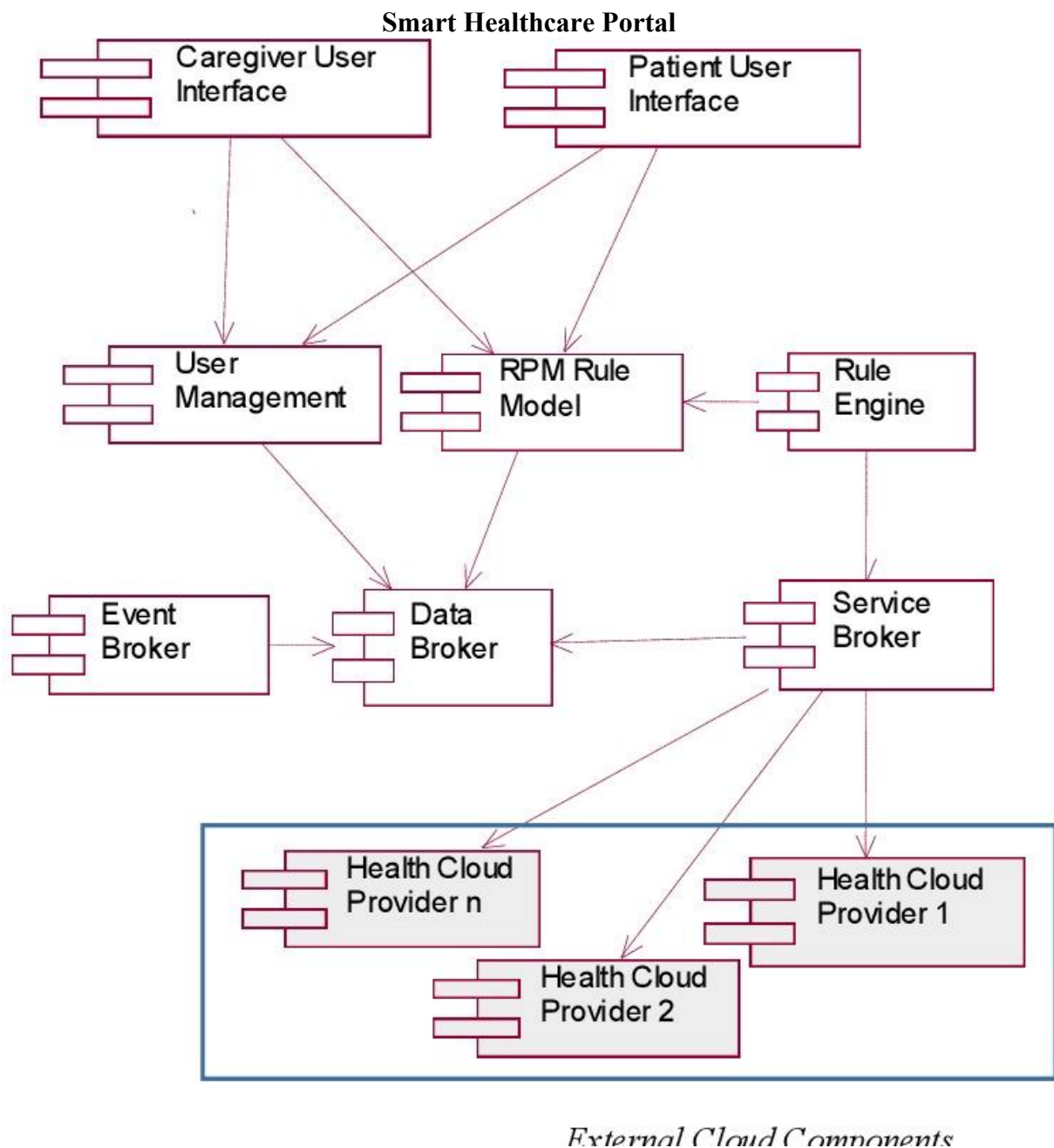
Traffic Signal Control System



5. Component Diagram Design

A Smart Healthcare Portal consists of modules such as Patient Management, Appointment Scheduling, Medical Records, Billing, and Notification Services. The current architecture does not clearly define component interactions and dependencies.

Analyse the system architecture, identify major components and their interactions, and design a Component Diagram showing dependencies and interfaces between modules.



6. Deployment Diagram Design

A Smart City Waste Management System is deployed across IoT waste-bin sensors, edge devices, cloud servers, and monitoring dashboards. The current design lacks clarity regarding physical architecture and communication links.

Analyse the deployment environment, identify nodes, artifacts, and communication links, and construct a detailed Deployment Diagram representing the complete system architecture.

Smart City Waste Management System

